Conservative and Non-Conservative Forces

Synopsis: You have been told twice now that gravity gives back what it takes and friction never does. Fair enough — but how would you know? Hand someone a force they had never met before and ask them which kind it is: what would they actually do to find out?

There is a test, and it is simple enough to run in your head. Take the object on a round trip and count the work. This Note runs that test, and then follows what it buys us: the reason height has an energy attached to it at all, a second force that passes the test, and an honest answer to where the joules friction takes actually end up.

Performance Expectations: Apply the round-trip test to decide whether a force is conservative. Explain why a potential energy can be defined for a conservative force and not for friction. Calculate the elastic potential energy stored in a spring. State where the energy removed by friction has gone.

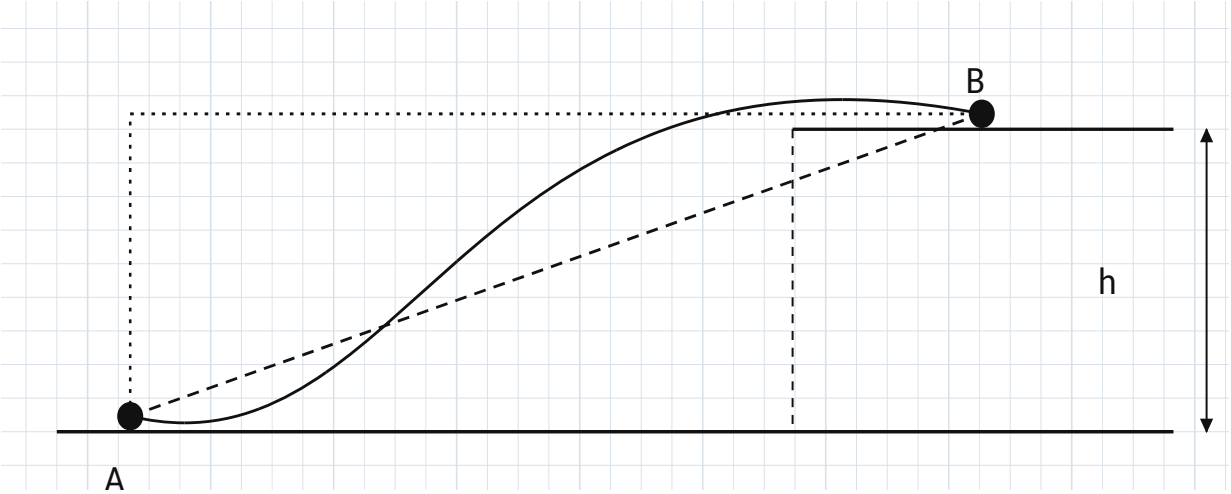
For PHY 151 — with calculus: State the closed-path condition for a conservative force, and recover the force from its potential energy function.

Instructions and Questions

1) The test: take it round a loop.
   A round trip is any journey that finishes exactly where it started — same place, same height. Over a whole round trip the object has gained nothing and lost nothing in position, so if a force is the kind that gives back what it takes, the work it did on the way out must be undone on the way home. The total should come to zero. If it does not come to zero, that force kept something. That is the whole test, and everything else in this Note follows from it.
   i) Say in your own words what has to be true at the end of a round trip for the test to apply.
   ii) Before we run it: predict the answer for gravity, and for friction.

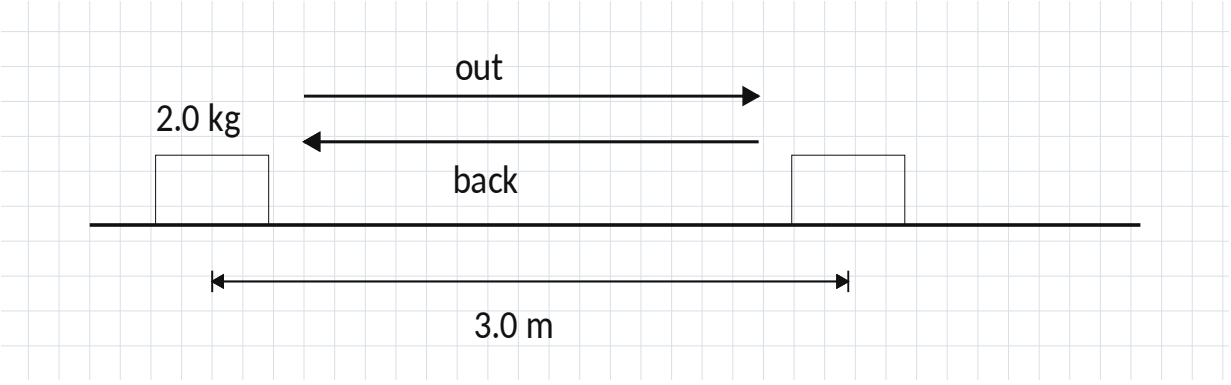
2) Gravity passes, and the route never matters.
   Here is the surprising part, and it is worth sitting with. Carry an object from A up to B and the work gravity does on it is −mgh — the same number whether you go straight up, up a long shallow ramp, or wander all over the hillside first. Gravity cannot see your route. It only sees how much higher you finished. So on a round trip, where you finish no higher at all, the work comes to exactly zero. A force like this is called conservative.

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i) A 2.0 kg book is carried from A to B, a height of 1.5 m. Calculate the work gravity does along the straight route. Take g = 9.8 m/s².
ii) Now calculate it along the other two routes. What do you get?
iii) The book is carried back down to A. How much work does gravity do over the whole round trip?
iv) One of the three routes is much longer than the others. Say why that makes no difference to your answer.

3) Friction fails, and the route is all that matters.
Now the same test on friction. Friction always points against the motion — that is its whole character — so when the object turns round and comes back, friction turns round with it and opposes the return trip too. It takes on the way out and it takes again on the way home. The round trip cannot come to zero; it comes to a negative number that gets bigger the further you travel. A force like this is called non-conservative.



i) A 2.0 kg block is pushed 3.0 m across a rough bench and then pushed straight back to where it started. Friction on it is 5.0 N throughout. Calculate the work friction does on the way out.
ii) Calculate the work friction does on the way back.

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iii) Add them. Is the round trip zero?

iv) Take the block the long way round instead — 6.0 m out and 6.0 m back. Now how much work does friction do?

v) Compare that with the round trip for gravity in section 2, and say what the difference is.

Gravity — round trip	Friction — round trip

4) This is why height has an energy attached to it.
We have been writing PE = mgh since Note 1 without asking what entitles us to. Here is the answer. Because gravity's work depends only on where you start and where you finish, you can label every height with a number and be sure the label is honest — the energy an object has at height h does not depend on how it got there. That label is the potential energy. A conservative force is exactly a force you are allowed to do this for, and $W_C = -\Delta PE$ is that permission written down.
Friction gets no such label. There is no "friction potential energy" you could write on the bench, because the energy lost depends on the path taken and not on where the block ended up. Two blocks sitting side by side on the same bench can have lost wildly different amounts. That is the real difference between the two kinds of force, and the round-trip test is how you tell which is which.

i) Explain, in a sentence a classmate would follow, why you can write PE = mgh for gravity but cannot write anything like it for friction.

ii) Two identical blocks sit at the same spot on the same bench. One was slid there directly, the other took a long detour. Do they have the same PE? Has the same amount of energy been lost to friction?

The closed-path condition, and getting the force back.
Written properly, the round-trip test is a closed line integral: a force is conservative when $\oint F \cdot dr = 0$ for every closed path. "Every" is doing real work in that sentence — one lucky loop is not enough. And the permission runs both ways: once a potential energy function U(x) exists, the force is recoverable from it as $F = -dU/dx$. The force is the slope of the potential, pointing downhill.

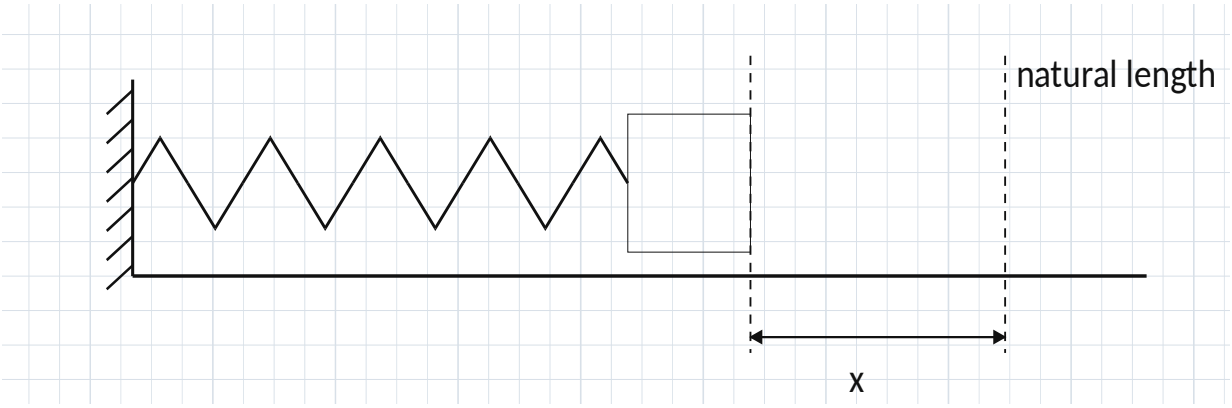
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i) Take $U(x) = \frac{1}{2}kx^2$ and differentiate. What force do you get, and which way does it point?

ii) Take $U(h) = mgh$ and do the same. Check the sign against what you know gravity does.

5) A second force that passes: the spring.

Gravity is not the only conservative force, and a spring is the one you can hold in your hand. Compress a spring and it pushes back with $F = kx$, where k is the spring constant in N/m and x is how far you have squashed it from its natural length. Let it go and it gives the energy straight back — round trip, zero, test passed. So a spring gets a potential energy of its own: $PE = \frac{1}{2}kx^2$. Note the square: squash it twice as far and it stores four times as much.



i) A spring with $k = 200$ N/m is compressed 0.15 m. How hard does it push back at that moment?

ii) How much energy is stored in it?

iii) The spring is released and pushes a 0.50 kg cart away along a smooth track. How fast does the cart leave?

iv) Compress the same spring 0.30 m instead — twice as far. How many times more energy is stored?

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6) Where friction's joules actually end up.
   Calling the energy "lost" or "wasted" is a bit of a dodge, because energy is conserved and it cannot have gone nowhere. It went into thermal energy — the block and the bench are both very slightly warmer than they were, and the warmth is spread through enormous numbers of jostling atoms. Nothing has been destroyed. What has happened is that the energy has been shared out so finely that you can no longer get hold of it to push anything with, and that is what makes friction non-conservative rather than merely inconvenient.
   i) The round trip across the bench in section 3 removed some joules from the block. Say where they are now.
   ii) Rub your hands together hard for a few seconds. Describe what you just did using W_{NC}.
   iii) Energy is conserved, yet we say mechanical energy is lost to friction. Explain why both statements are true.

7) Looking ahead.
   You now have the two kinds of force and the test that separates them, so every energy problem from here splits the same way: the conservative part becomes a change in potential energy, and whatever is left over is W_{NC}. Next we put a clock on it — not how much energy moved, but how fast.

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